

Objective:

This assignment has been designed for students to apply the appropriate concurrent program

Methods for implementing a concurrent program from a program specification.

Learning Outcomes

ON COMPLETION OF THIS ASSIGNMENT, YOU SHOULD BE ABLE TO DEMONSTRATE THE FOLLOWING LEARNING OUTCOME(S):

No.	Learning Outcome	Assessment
1 2	Explain the fundamental concepts of concurrency and parallelism in the design of a concurrent system (C2, PLO1)	Exam
3	Apply the concepts of concurrency and parallelism in the construction of a system using a suitable programming language. (C3, PLO2)	Individual Assignment (System)
	Explain the safety aspects of multi-threaded and parallel systems (A3, PLO6)	Individual Assignment (Report)

Programme Outcomes (PO):

PLO2 Cognitive Skills - This relates to thinking or intellectual capabilities and the ability to apply knowledge and skills. The capacity to develop levels of intellectual skills progressively begins from understanding, critical/creative thinking, assessment, and applying, analysing, problem solving as well as synthesizing to create new ideas, solutions, strategies, or new practices. Such intellectual skills enable the learner to search and comprehend new information from different fields of knowledge and practices.

Individual Assignment - Report(20%):

Question No.	Topic	Question Vs Taxonomy					PLO
		Affective Level					
		1	2	3	4	5	
		SQ	SQ	SQ	SQ	SQ	
1	Introduction and background			20%			6
2	Explanation of the safety aspects of multi-threaded system implemented			30%			6
3	Justification of coding techniques implemented.			30%			6
4	Depth of discussion of concurrency concepts.			20%			6
	Total			100%			24

Individual Assignment - System (25%):

Question No.	Topic	Question Vs Taxonomy						PLO
		Cognitive Level						
		1	2	3	4	5	6	
		SQ	\$Q	SQ	SQ	SQ	SQ	
1	Appropriateness of coding techniques used to implement design with appropriate comment lines in source codes			20%				2
2	Appropriateness of the Java concurrent programming facilities used.			20%				2
3	Program runs appropriately with basic requirements.			20%				2
4	Additional requirements met			20%				2
5	Explanations of concurrency concepts implemented with relevant code samples.			20%				2
	Total			100%				

Submission Requirements:

Assignment Handout Date **16th March 2026**

Assignment Due Date **16th May 2026**

Case Study

Asia Pacific Airport

You have been tasked to simulate the operations of the airport.

*******Basic Requirements*******

- There is only 1 runway for all planes to land and depart.
- There can only be 3 airplanes on the airport grounds, including the runway. This is to ensure that the aircraft does not collide with another aircraft on the runway or gates.
- Once an aircraft obtains permission to land, it should land on the runway, coast to the assigned gate, dock to the gate, allow passengers to disembark, refill supplies and fuel, receive new passengers, undock, coast to the assigned runway and take-off.
- Each step should take some time. As the airport is small, there is no waiting area on the ground for the planes to wait for a gate to become available.

- These events should happen concurrently:
 - Passengers disembarking/embarking from the 3 gates.
 - Refill Food/supplies and cleaning of aircraft
 - Refueling of aircraft

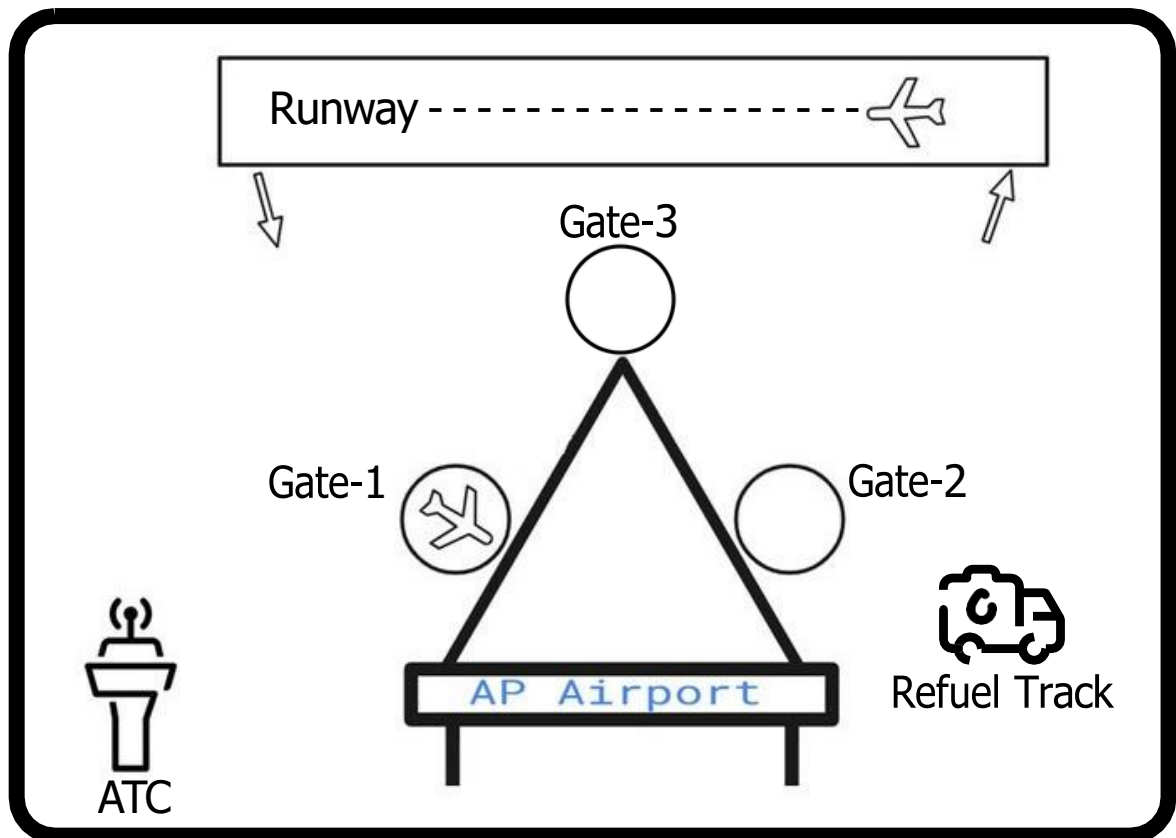


Figure 1: Layout of the Asia Pacific Airport (Not to scale)

- The Statistics Report:

At the end of the simulation, i.e., when all planes have left the airport, the ATC manager should do some sanity checks of the airport and print out some statistics on the run. The result of the sanity checks must be printed. You must
- Check that all gates are indeed empty.
- Print out statistics on
 - Maximum/Average/Minimum waiting time for a plane.
 - Number of planes served/Passengers boarded.

*******Additional Requirements*******

- There is only 1 refueling truck to perform:
 - Refueling of aircraft
 - A congested scenario should be simulated where 2 planes are waiting to land while the 2 gates are occupied, and a 3rd plane comes in with fuel shortage, requiring emergency landing.

NOTE: State your assumptions and how you will implement them.

Deliverables:

For this exercise, you are required to model the ATC scenario and design a Java program to simulate the Airport activities:

- Altogether, 6 planes should try to land at the airport.
 - Use a random number generator, so a new airplane arrives every 0, 1, or 2 seconds. (This might be accomplished by an appropriate statement `sleep (rand.nextInt(2000));`)
 - Assume each plane can accommodate a maximum of 50 passengers.
 - Assume passengers are always ready to embark and disembark the terminal (i.e., no capacity issues inside the passenger terminals)
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Sample Output:

To see what is happening dynamically, you must have a single output from every plane's passengers, the air traffic controller, and the pilots reporting all their major events. As shown in Figure 2 below:

```
Plane-1: Requesting Landing.
ATC: Landing permission granted for Plane-1.
ATC: Gate-1assignedfor Plane-1.
Plane-1: Landing.
Plane-1: Landed.
Plane-2: Requesting Landing.
Plane-1: CoastingtoGate-1.
ATC: Landing permission granted for Plane-2.
Plane-2: Landing.
Plane-1: DockedatGate-1.
Plane-1's Passengers: 47 embarking out of Plane-1.

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.
Plane-4: Requesting Landing.
ATC: Landing Permission Denied for Plane-4, Airport Full.
Plane-1: Requesting Taking off.
ATC: Taking-off is granted for Plane-1. Runway is free.
Plane-1: Taking off.
```

Figure 2: Correct / Desired Output

Add information about which process/thread is doing the output. This way you can see if a process/thread acts for another, which is strictly forbidden, but is a common error for Java solutions (objects are not processes!). An example of such incorrect behaviour, see Figure 3 below:

```
Thread-ATC : Plane 5: Requesting permission to land!
MainThread : ATC: Please wait and join the circle queue.
Thread-Passenger-7 : I'm boarding Plane 2 now.
Thread-Passenger-8 : I'm boarding Plane 2 now.
Thread-Passenger-9 : I'm boarding Plane 2 now.
```



Figure 3: Incorrect / Undesired Output

Where we can see that the ATC's thread is acting for Plane 5, and also the main thread is acting for the ATC. Additionally, we can see that Each passenger is acting individually.

Implementation:

- You should implement your simulation in Java.
- The simulation run should not take more than 60 seconds to simulate

Documentation for System (Report)

The documentation should detail the system implementation and testing.

1. Basic requirements met:

List of requirements met.

- Short explanation of concurrency concepts (atomic statements, synchronization, etc) implemented.

Code snippet of the Java concurrent programming facilities implemented.

2. Additional requirements met:

List of requirements met.

- Short explanation of concurrency concepts (atomic statements, synchronization, etc) implemented.

Code snippet of the Java concurrent programming facilities implemented.

3. Requirements which were NOT met:

List of basic requirements.

List of additional requirements

NOTE: if all the requirements were fulfilled, then just write 'ALL Requirements were Fulfilled'

The document's length should be around 1500 words excluding References/appendix/coding.

Submission for System

Please Upload The following TWO FILES Only:

1- Full Report Document

2- A SINGLE zip file named TP0XXXXX CCP.zip that contains:

a. Video Presentation of the simulation running. Minimum 3 minutes and

Maximum 5 minutes per person.

i. Simulate scenarios as stated above.

ii. Show which requirements are met in the output and corresponding code.

b. Fully Functional Code's Folder: Java files required to run the simulation.

Marking Scheme (NOT for student's documentation guidance)

Report (20%)		
Criteria	Total marks	Marks awarded
Assumptions [LO3-PO6]	20	
Explanation of the safety aspects of multi-threaded system implemented [LO3-PO6]	30	
Justification of coding techniques implemented [LO3-PO6]	30	
TOTAL MARKS	100	
Depth of discussion of concurrency concepts [LO3-PO6]	20	

System (25%)

Criteria	Total marks	Marks awarded
Appropriateness of coding techniques used to implement design with appropriate comment lines in source codes *	20	
Appropriateness of the Java concurrent programming facilities used.	20	
Program runs appropriately with basic requirements *	20	
Additional requirements met *	20	
Explanations of concurrency concepts implemented with relevant code samples *	20	
TOTAL MARKS	100	

IMPORTANT: Failure to attend the final presentation for the assignment will be deemed as non-attendance of the final assessment for this assignment item, resulting in a score of ZERO (0).

IMPORTANT: Students are strictly prohibited from using any Java libraries that handle thread concurrency automatically. All concurrency-related implementations must be explicitly written by the students, without relying on built-in parallel execution mechanisms.

Restricted Libraries and Features

The following Java libraries and features are not allowed due to their automatic thread concurrency handling:

- java.util.concurrent.ExecutorService
- java.util.concurrent.ForkJoinPool
- java.util.concurrent.CompletableFuture
- ParallelStream in Java Streams API

- `java.util.Timer` (if used for scheduled execution)
- Reactive Streams (e.g., Project Reactor, RxJava)
- Spring `@Async` annotation —
- Vert.x (if used for automatic event-loop concurrency)
`java.util.concurrent.PriorityBlockingQueue`
- ... and any other package provides concurrency services other than those taught in the class.

Any submission using these restricted libraries will be considered non-compliant and subject to penalties. If unsure about a library, consult the instructor before using it.

IMPORTANT: Please avoid the followings (DON'T-DO-LIST):

- It is imperative to avoid the direct verbatim duplication of source material into the knowledge base. All information must be properly synthesized and represented according to the previously established guidelines.
 - Presence at the final Individual presentation is a compulsory requirement for assessment. Any individual who fails to attend the scheduled session will be ineligible for credit, resulting in a non-negotiable grade of ZERO (0) for the assignment component.
 - Completion and submission of all assignment components are compulsory. Non-compliance with the submission requirements for any constituent part may result in a mandatory penalty, including the forfeiture of all marks for that specific assignment component (resulting in a grade of ZERO (0)).
 - Submissions received after the designated deadline will be subject to a mandatory assessment penalty. The magnitude of the deduction shall be determined at the discretion of the presiding lecturer, in accordance with the department's late-work protocols.
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